Chiral relations for entanglement in quantum self-phase modulation

M. Gil de Oliveira[1,2], A. Bramati[3], A. Z. Khoury[1]
1- Instituto de Física, Universidade Federal Fluminense, 24210-346 Niterói, RJ, Brazil
2 - Quantum Industrial Innovation, Competence Center in Quantum Technologies, Universidade SENAI-CIMATEC, 40650-010, Salvador, BA, Brazil and
3- Laboratoire Kastler Brossel, Paris, France
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We investigate entanglement certification in the radial mode structure generated by self-phase modulation of a coherent beam carrying orbital angular momentum (OAM). We derive a chiral selection rule for the first-order anomalous correlations: at specified detection waists, the relative signs and magnitudes of the OAM values determine a finite set of radial partners for a reference mode. Nonzero coefficients certify pairwise entanglement and full inseparability of the selected set at sufficiently short propagation distances. Finite-distance calculations within the diffraction-free Bogoliubov approximation show that this first-order support continues to organize the certification pattern. Optimized covariance witnesses reveal entanglement involving modes outside that support, with smaller violation margins in the configurations investigated, and certify genuine multipartite entanglement in reduced sets of detection modes. These results identify radial–angular coupling as a means of controlling which spatial mode combinations admit entanglement certification.

I. INTRODUCTION

The quest for quantum devices based on optical resources is an intense and active field of research, which has already delivered important tools for quantum communication, computing and metrology. This quest has increased considerably by the possibility of shaping light in its different degrees of freedom, such as time envelope, spectral and spatial modes [1, 2]. Both discrete and continuous variable quantum information processing can benefit from these developments [3–5]. Nonlinear optical processes are the common tool for generating entangled quantum states, as it allows the energy transfer between different optical modes, constrained by different conditions such as energy conservation and phase matching, for example [6–10]. These conditions constrain the mode couplings through which quantum correlations and entanglement can develop [11–14]. Energy and momentum conservation are behind the generation of time-frequency and position-momentum entanglement, respectively. These conditions rule nonlinear optical processes such as three- and four-wave mixing, second harmonic generation and self-phase modulation [15–25].

Entanglement provides a resource for quantum communication protocols involving two or more parties [26]. In a spatially structured optical field, orthogonal modes provide distinct subsystems whose quadratures can be addressed by mode-selective detection. Controlling how a nonlinear interaction distributes entanglement among these modes is therefore a step toward preparing spatial resources for quantum information. This requires identifying both the modes involved and the form of entanglement that can be certified: correlations between pairs, entanglement across groups of modes, and genuine multipartite entanglement are different statements about the generated state [27].

The possibility of shaping the phase and polarization distributions with spatial light modulators and q-plates has become a powerful tool for controlled generation of customized entangled states both in linear and nonlinear optics. These developments brought the orbital angular momentum (OAM) of light to the toolbox of quantum information technology. Its role in nonlinear processes has also been unveiled by a plethora of scientific papers devoted to the constraints imposed by OAM and vortex formation in nonlinear wave mixing. In polariton superfluids, the interplay between phase, spin and flow gives rise to a rich variety of spatial structures, including quantized vortices, spin textures and vortex flows [28–32]. A recent work has further demonstrated the spectroscopy of collective excitations associated with stationary, multiply quantized vortices, opening new possibilities for exploring vortex dynamics in driven-dissipative quantum fluids of light [32].

A less exploited degree of freedom involves the radial structure of paraxial modes [8, 15, 16, 33–35]. The Laguerre-Gaussian (LG) solutions of the paraxial wave equation are the natural basis for describing OAM and the radial structure, which are the suitable labels for their identification. These modes are characterized by two integer numbers, namely its topological charge and radial index. The topological charge corresponds to the azimuthal variation of the beam phase around the propagation direction and the radial index determines the number of intensity rings. While the OAM conservation selection rule has been demonstrated in different quantum optical processes, the constraints related to the radial structure still deserves careful investigation.

Self-phase modulation is a particularly interesting phenomenon, where the intensity of an optical beam alters the effective refraction index of a nonlinear medium, thus affecting the beam's phase evolution along propagation through the medium [36]. This effect can also be observed in Bose-Einstein condensates and atomic vapors [37–39]. In a previous work we investigated the formation of a radial structure in self-phase modulation driven

by an OAM beam and determined the radial spectrum generated in terms of the topological charge of the input beam. This established an interesting radial-angular coupling promoted by the radial selection rules imposed on the nonlinear process [40].

In the present work we derive a radial–angular selection rule for entanglement certification in self-phase modulation. For a coherent LG pump and specified detection waists, the first-order anomalous correlations have finite radial support determined by the relative chirality of the interacting modes. Their nonzero coefficients guarantee two-mode entanglement at sufficiently short distances and establish full inseparability of the selected set. We then investigate finite propagation distances using the diffraction-free Bogoliubov approximation and optimized covariance witnesses. In the configurations studied, the first-order support continues to distinguish larger certification margins from the weaker violations involving additional radial modes.

Section II introduces the spatial mode operators and covariance matrix. Section III defines the parties, separability classes, and entanglement criteria used throughout the paper. Section IV derives the first-order correlations from the full fluctuation equation, and Sec. V establishes their chiral selection rule and its consequences for certification. Section VI examines the finite-distance behavior and genuine multipartite entanglement of reduced sets of detection modes.

II. SPATIALLY STRUCTURED BOSONIC FIELD

We first introduce the spatial mode operators and correlations used to describe the field. This formulation identifies the optical subsystems and provides the covariance matrix on which the entanglement criteria of Sec. III act.

We start by considering the normalized positive frequency component of the quantized electromagnetic field $\mathcal{E}(\mathbf{r})$ satisfying the commutation relations

$$\begin{aligned}\left[\mathcal{E}(\mathbf{r}_1),\mathcal{E}^\dagger(\mathbf{r}_2)\right] &= \delta^{(2)}(\mathbf{r}_1-\mathbf{r}_2),\\ \left[\mathcal{E}(\mathbf{r}_1),\mathcal{E}(\mathbf{r}_2)\right] &= \left[\mathcal{E}^\dagger(\mathbf{r}_1),\mathcal{E}^\dagger(\mathbf{r}_2)\right] = 0.\end{aligned} \tag{1}$$

In the above equation and in what follows, $\mathbf{r} = (x, y)$ is the transverse coordinate. One can introduce ladder operators $a(v), a^\dagger(v)$ associated with the spatial mode v by the expression

$$a(v) = \int_{\mathbb{R}^2} v^*(\mathbf{r})\mathcal{E}(\mathbf{r})d\mathbf{r}. \tag{2}$$

We then have

$$\begin{aligned}\left[a(v_1),a^\dagger(v_2)\right] &= (v_1;v_2) \equiv \int_{\mathbb{R}^2} v_1^*(\mathbf{r})v_2(\mathbf{r})d\mathbf{r}\,,\\ \left[a(v_1),a(v_2)\right] &= \left[a^\dagger(v_1),a^\dagger(v_2)\right] = 0\,.\end{aligned} \tag{3}$$

For orthonormal profiles, $(v_j; v_k) = \delta_{jk}$, these are independent bosonic mode operators. Each mode has its own Fock space, and a set of such modes defines a tensor product of subsystems. Orthogonality does not require spatially separated intensity profiles: distinct modes can overlap within the same beam. The operator construction also applies to nonorthogonal or unnormalized detection functions, but these do not by themselves define independent parties for an entanglement test. All mode parties considered below are defined by orthonormal profiles.

Given a classical field distribution $\alpha(\mathbf{r})$ normalized in such a way that $|\alpha(\mathbf{r})|^2$ represents the photon number per unit area, we may build a coherent state through the expression

$$|\alpha\rangle = \exp\left[a^\dagger(\alpha) - a(\alpha)\right]|0\rangle\,, \tag{4}$$

where $|0\rangle$ is the vacuum state. This state is an eigenvector of the field operator,

$$\mathcal{E}(\mathbf{r})\,|\alpha\rangle = \alpha(\mathbf{r})\,|\alpha\rangle\,. \tag{5}$$

This is the type of state that will be used as input on the nonlinear medium.

The field quadratures are defined by

$$X(v) = \frac{a(v) + a^\dagger(v)}{\sqrt{2}}\,. \tag{6}$$

Here, v is the corresponding field mode. The complementary quadrature is simply defined by shifting v by a $\pi/2$ phase: $P(v) = X(iv)$.

The covariance function is given by

$$\Gamma(v_1, v_2) = \langle\{\Delta X(v_1), \Delta X(v_2)\}\rangle\,, \tag{7}$$

where $\Delta X(v) = X(v) - \langle X(v)\rangle$ and $\{\bullet,\bullet\}$ denotes the anticommutator. The calculation of this observable is simplified if we express it in terms of the field $\mathcal{E}$. For this, we introduce the normal and anomalous correlations, given respectively by

$$H(\mathbf{r}_1,\mathbf{r}_2) = \left\langle\mathcal{E}^\dagger(\mathbf{r}_2)\mathcal{E}(\mathbf{r}_1)\right\rangle - \left\langle\mathcal{E}^\dagger(\mathbf{r}_2)\right\rangle\langle\mathcal{E}(\mathbf{r}_1)\rangle\,, \tag{8}$$

$$F(\mathbf{r}_1,\mathbf{r}_2) = \langle\mathcal{E}(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z)\rangle - \langle\mathcal{E}(\mathbf{r}_1)\rangle\langle\mathcal{E}(\mathbf{r}_2)\rangle\,. \tag{9}$$

We may then write

$$\Gamma(v_1,v_2) = \mathrm{Re}\left[(v_1;v_2) + 2f(v_1,v_2) + 2h(v_1,v_2)\right], \tag{10}$$

where

$$\begin{aligned}h(v_1,v_2) &= \int v_1^*(\mathbf{r}_1)v_2(\mathbf{r}_2)H(\mathbf{r}_1,\mathbf{r}_2)\,d\mathbf{r}_1\,d\mathbf{r}_2\\ &= \left\langle a^\dagger(v_2)a(v_1)\right\rangle - \langle a(v_2)\rangle^*\langle a(v_1)\rangle\,,\end{aligned} \tag{11}$$

and

$$\begin{aligned}f(v_1,v_2) &= \int v_1^*(\mathbf{r}_2)v_2^*(\mathbf{r}_1)F(\mathbf{r}_1,\mathbf{r}_2)\,d\mathbf{r}_1\,d\mathbf{r}_2\\ &= \langle a(v_1)a(v_2)\rangle - \langle a(v_1)\rangle\langle a(v_2)\rangle\,.\end{aligned} \tag{12}$$

We further note the following properties, which simplify the consideration of linear combination of modes:

$$\begin{aligned} f(v_1, v_2 + \lambda v_3) &= f(v_1, v_2) + \lambda^* f(v_1, v_3); \\ f(v_1, v_2) &= f(v_2, v_1); \\ h(v_1, v_2 + \lambda v_3) &= h(v_1, v_2) + \lambda h(v_1, v_3); \\ h(v_1, v_2) &= h(v_2, v_1)^*. \end{aligned} \tag{13}$$

These are a reflection of the fact that $a(v)$ is antilinear while $a^\dagger(v)$ is linear.

For N orthonormal modes, define the quadrature vector and covariance matrix by

$$\begin{aligned} \mathbf{R} &= \big(X(v_1), P(v_1), \ldots, X(v_N), P(v_N)\big), \\ \gamma_{jk} &= \langle \{\Delta R_j, \Delta R_k\} \rangle . \end{aligned} \tag{14}$$

Every entry follows from Eq. (10), using $P(v) = X(iv)$. With our normalization, $[X(v_j), P(v_k)] = i\delta_{jk}$, the vacuum variance of each quadrature is $1/2$, and the vacuum covariance matrix is $\mathbb{1}_{2N}$. We express quadrature variances in shot-noise units (SNU), defining $V_{\mathrm{SNU}}(Q) = 2\left\langle (\Delta Q)^2 \right\rangle$, so that vacuum noise is one in these units.

III. ENTANGLEMENT AND COVARIANCE-BASED CERTIFICATION

A. Parties and separability

We take the parties to be specified orthogonal modes or disjoint groups of modes. Radial and OAM indices label the spatial profiles; the quantum subsystems are their associated Fock spaces. A bipartition $S|T$ divides the parties S and T into two nonempty groups. A state ρ is separable across that cut if it admits a decomposition

$$\rho = \sum_\lambda p_\lambda \rho_S^{(\lambda)} \otimes \rho_T^{(\lambda)}, \qquad p_\lambda \geq 0, \quad \sum_\lambda p_\lambda = 1. \tag{15}$$

Here, $\rho_S^{(\lambda)}$ is a density operator belonging to party S, with an analogous convention for T. Otherwise, ρ is entangled across $S|T$.

A state is fully inseparable if it is entangled across every nontrivial bipartition of the specified parties. For mixed states, this does not exclude a biseparable decomposition

$$\rho_{\mathrm{bisep}} = \sum_\lambda p_\lambda \rho_{S_\lambda|T_\lambda}, \tag{16}$$

where each component is separable as in Eq. (15), but the cut may vary with λ. Genuine multipartite entanglement (GME) excludes all such decompositions [27]. Physically, GME rules out preparing the state by randomly choosing which subgroups share entanglement. Full inseparability and GME coincide for pure states; they are distinct requirements for mixed states.

B. Covariance based entanglement witnesses

The correlations introduced in Sec. II allow entanglement to be certified from quadrature measurements. Following Hyllus and Eisert [41], a covariance witness for a specified separability class is a real, symmetric, positive semidefinite matrix $\mathcal{Z}$ satisfying

$$\mathrm{Tr}(\mathcal{Z}\gamma_{\mathrm{sep}}) \geq 1, \tag{17}$$

for every covariance matrix of a state in that class. Consequently,

$$\mathrm{Tr}(\mathcal{Z}\gamma) < 1, \tag{18}$$

excludes that class for any state with covariance matrix γ. Choosing the class of states separable across a fixed cut tests entanglement across that cut; choosing the entire biseparable class tests GME. A value at or above one is inconclusive. The margin below one measures the violation of the chosen criterion and is not, by itself, an entanglement measure.

C. Two-mode Duan criterion

For two orthonormal modes v_1, v_2, the Duan criterion [42, 43] certifies entanglement of their reduced state when

$$\begin{aligned} D = \frac{1}{2}\left\langle [\Delta X(v_1) + \Delta X(v_2)]^2 \right\rangle \\ + \frac{1}{2}\left\langle [\Delta P(v_1) - \Delta P(v_2)]^2 \right\rangle < 1. \end{aligned} \tag{19}$$

In the quadrature ordering of Eq. (14), this is $D = \mathrm{Tr}(\mathcal{Z}_D\gamma)$ with

$$\mathcal{Z}_D = \frac{1}{4}\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}. \tag{20}$$

Equation (10) gives the useful expression

$$D = 1 + h(v_1, v_1) + h(v_2, v_2) + 2\,\mathrm{Re}\, f(v_1, v_2). \tag{21}$$

A phase rotation $v_1 \mapsto e^{i\phi} v_1$, with $\phi = \arg f(v_1, v_2) \pm \pi$, minimizes this quantity:

$$D_{\mathrm{opt}} = 1 + h(v_1, v_1) + h(v_2, v_2) - 2|f(v_1, v_2)|. \tag{22}$$

We refer to D_{opt} as the phase-optimized Duan witness. This optimization changes only the quadrature phases. It does not optimize over all covariance witnesses. Its simple dependence on the anomalous correlation makes it particularly suitable for deriving the short-distance selection rule below.

D. Multimode certification and optimized witnesses

Pairwise tests also constrain multimode separability: an entangled pair crossing a cut excludes separability across that cut, since tracing out other modes preserves separability. A collection of pairwise violations that crosses every bipartition therefore establishes full inseparability. This reasoning underlies the pairwise version of the criteria discussed by van Loock and Furusawa [26]; we use it below for a reference mode coupled to several radial partners. For mixed states, the additional exclusion of Eq. (16) is needed to establish GME [27].

The van Loock–Furusawa criteria also allow quadratures of additional modes to enter with adjustable gains [26]. These collective observables can reveal entanglement across cuts when the reduced pairs fail the Duan test. More generally, Hyllus and Eisert optimize over covariance witnesses using semidefinite programming [41].

We use two such optimizations. For each specified bipartition, the FullyWit routine minimizes $\mathrm{Tr}(\mathcal{Z}\gamma)$ over witnesses valid for that cut. Violations for every cut establish full inseparability. The MultiWit routine instead imposes validity for all biseparable states, so a violation certifies GME of the selected parties. Both use the covariance matrix of the selected modes after the remaining field has been traced out. These established criteria provide the framework for interpreting the radial–angular coupling derived next.

IV. KERR DYNAMICS AND FIRST-ORDER CORRELATIONS

We consider a monochromatic field propagating through a $\chi^{(3)}$ nonlinear medium, with a coherent spatial mode as input. The nonlinear interaction produces both a change in the mean field and correlations between mode fluctuations. We now calculate these correlations to first order in propagation distance and apply the criteria of Sec. III.

We will assume that the field is propagating along the z direction, and satisfies the Heisenberg equation of motion

$$2ik\partial_z\mathcal{E}(\mathbf{r},z)+\nabla^2\mathcal{E}(\mathbf{r},z)=g\,\mathcal{E}^\dagger(\mathbf{r},z)\mathcal{E}^2(\mathbf{r},z)\,, \tag{23}$$

where g is the Kerr nonlinear coupling constant and ∇^2 denotes the transverse Laplacian. The short-distance expansion below retains the full fluctuation equation and yields the exact coefficient linear in distance. In Sec. VI, we adopt a different approximation: diffraction-free, linearized fluctuation dynamics with the full distance dependence retained. This allows us to investigate how the first-order certification pattern extends to finite propagation distances.

To start our derivation, let $\alpha_c(\mathbf{r},z)$ denote the solution of the classical equations of motion

$$2ik\partial_z\alpha_c+\nabla^2\alpha_c=g\,|\alpha_c|^2\alpha_c, \tag{24}$$

under the initial condition $\alpha_c(\mathbf{r},0)=\alpha(\mathbf{r})$, where $|\alpha\rangle$ is the input coherent state. Let us also define the fluctuation field

$$\eta(\mathbf{r},z)=\mathcal{E}(\mathbf{r},z)-\alpha_c(\mathbf{r},z). \tag{25}$$

We observe that our initial state has vacuum like properties with respect to $\eta(\mathbf{r},0)$. For example, it is annihilated by it:

$$\eta(\mathbf{r},0)\,|\alpha\rangle=0. \tag{26}$$

Furthermore, any normally ordered monomial containing at least one fluctuation operator has zero expectation value at $z=0$.

The correlation functions, which are our observables of interest, can also be compactly written in terms of η:

$$\begin{aligned}F(\mathbf{r}_1,\mathbf{r}_2;z)&=\langle\eta(\mathbf{r}_1,z)\eta(\mathbf{r}_2,z)\rangle-\langle\eta(\mathbf{r}_1,z)\rangle\langle\eta(\mathbf{r}_2,z)\rangle,\\ H(\mathbf{r}_1,\mathbf{r}_2;z)&=\langle\eta^\dagger(\mathbf{r}_2,z)\eta(\mathbf{r}_1,z)\rangle-\langle\eta(\mathbf{r}_1,z)\rangle\langle\eta^\dagger(\mathbf{r}_2,z)\rangle.\end{aligned} \tag{27}$$

Next, we write the equation of motion

$$2ik\partial_z\eta+\nabla^2\eta=g\left(2|\alpha_c|^2\eta+\alpha_c^2\eta^\dagger+\Omega\right), \tag{28}$$

where the nonlinear fluctuation terms are

$$\Omega=\alpha_c^*\eta^2+2\alpha_c\eta^\dagger\eta+\eta^\dagger\eta^2. \tag{29}$$

We start by considering an approximation where only terms up to $\mathcal{O}(z)$ are retained. From definition, we already have $\langle\eta(\mathbf{r},0)\rangle=0$. Furthermore, as the equation of motion (28) contains only normally ordered terms, we also get $\partial_z\langle\eta(\mathbf{r},0)\rangle=0$, such that

$$\langle\eta(\mathbf{r},z)\rangle=\mathcal{O}(z^2), \tag{30}$$

or, equivalently,

$$\langle\mathcal{E}(\mathbf{r},z)\rangle=\alpha_c(\mathbf{r},z)+\mathcal{O}(z^2). \tag{31}$$

Now we turn to the correlation functions. For compactness, we write $\eta_{j,z}=\eta(\mathbf{r}_j,z)$. In order to calculate the normal correlation, we differentiate the two-operator average:

$$\partial_z\langle\eta_2^\dagger\eta_1\rangle\Big|_0=\langle(\partial_z\eta_2^\dagger)_0\eta_{0,1}\rangle+\langle\eta_{0,2}^\dagger(\partial_z\eta_1)_0\rangle. \tag{32}$$

Both terms vanish: the first average has an annihilation operator on the right, and the second has its conjugate on the left.

Turning to the anomalous case, we have that

$$\begin{aligned}\partial_z\langle\eta_{1,z}\eta_{2,z}\rangle|_0&=\langle(\partial_z\eta_1)_0\eta_{2,0}\rangle+\langle\eta_{1,0}(\partial_z\eta_2)_0\rangle\\&=\langle\eta_{1,0}(\partial_z\eta_2)_0\rangle.\end{aligned} \tag{33}$$

Isolating $\partial_z\eta$ in the equations of motion (28) and substituting above, we see that almost all the terms are zero, because they contain η to the right. The only exception is the term proportional to $\eta^\dagger$, which gives

$$\partial_z\langle\eta_1\eta_2\rangle|_0=-\frac{ig}{2k}\alpha^2(\mathbf{r}_2)\delta^{(2)}(\mathbf{r}_1-\mathbf{r}_2). \tag{34}$$

Since the mean-subtraction terms do not contribute at first order, the correlations are

$$F(\mathbf{r}_1,\mathbf{r}_2;z) = -\frac{igz}{2k}\alpha^2(\mathbf{r}_2)\delta^{(2)}(\mathbf{r}_1-\mathbf{r}_2)+\mathcal{O}(z^2),$$
$$H(\mathbf{r}_1,\mathbf{r}_2;z) = \mathcal{O}(z^2). \tag{35}$$

Using the spatial overlap notation

$$(v_1,\ldots,v_n;u_1,\ldots,u_n) = \int d^2\mathbf{r}\prod_{j=1}^{n} v_j^*(\mathbf{r})u_j(\mathbf{r}), \tag{36}$$

projection onto the modes v_1 and v_2 gives

$$f(v_1,v_2) = -\frac{igz}{2k}(v_1,v_2;\alpha,\alpha)+\mathcal{O}(z^2),$$
$$h(v_1,v_2) = \mathcal{O}(z^2). \tag{37}$$

To specify the input and introduce a natural transverse scale, we consider coherent Laguerre-Gaussian modes, with

$$\alpha(\mathbf{r}) = \|\alpha\|\, LG_{pl}(\mathbf{r};w), \tag{38}$$

where the waist-plane modes are

$$LG_{pl}(\mathbf{r};w) = \frac{(-1)^p\mathcal{N}_{pl}}{w}\left(\frac{2r^2}{w^2}\right)^{|l|/2} \times L_p^{|l|}\left(\frac{2r^2}{w^2}\right)e^{-r^2/w^2+il\phi}, \tag{39}$$

with radial order p, topological charge l, and normalization constant

$$\mathcal{N}_{pl} = \sqrt{\frac{2}{\pi}}\sqrt{\frac{p!}{(p+|l|)!}}\,. \tag{40}$$

The waist w sets the natural transverse scale. Let us define the Rayleigh range $z_R = kw^2/2$, the nonlinear interaction length $z_{NL} = 4z_R/(g\|\alpha\|^2)$, the dimensionless propagation distance $Z = z/z_{NL} = gz\|\alpha\|^2/(2kw^2)$, and the dimensionless input profile $u(\mathbf{r}) = w\alpha(\mathbf{r},0)/\|\alpha\|$.

Using these definitions, Eq. (10) and the Duan criterion give the first-order results

$$f(v_1,v_2) \approx -iZ(v_1,v_2;u,u),$$
$$\Gamma(v_1,v_2) = \mathrm{Re}(v_1;v_2)+2Z\,\mathrm{Im}(v_1,v_2;u,u)+\mathcal{O}(Z^2),$$
$$D_{\mathrm{opt}} = 1-2\,|Z(v_1,v_2;u,u)|+\mathcal{O}(Z^2). \tag{41}$$

For an input angular momentum l_0, the angular integration requires $l_{v_1}+l_{v_2}=2l_0$ for a linear Duan violation. This OAM condition is necessary for first-order certification; the corresponding radial overlap must also be nonzero.

V. CHIRAL SELECTION RULE AND ENTANGLEMENT CERTIFICATION

We now specialize to the LG input $\alpha(\mathbf{r}) = \|\alpha\|\,LG_{0,l_0}(\mathbf{r};w)$ and analyze the selection rules for radial and angular mode coupling.

Our previous work [40] has shown that, in the classical regime, the first order treatment gives a selection rule for the generation of radial orders up to $|l_0|$. The current quantum treatment clarifies that this was, in fact, a selection rule for the mean field value $\langle\mathcal{E}\rangle$ given by (31), whose expression had already appeared on our earlier treatment.

We now derive the radial selection rule for the first-order anomalous correlations and use it to establish pairwise and multimode entanglement at sufficiently small nonzero distances.

A. Radial selection rules and two-mode entanglement

Let $v_2 = LG_{0,l_2}(\mathbf{r};w_2)$ be our reference mode, where w_2 is free and l_2 is only constrained by $l_2\neq l_0$. Now, define $l_1 = 2l_0-l_2$ and $1/w_1^2 = 2/w^2+1/w_2^2$. We then have that

$$\begin{aligned} v_2^*(\mathbf{r})u^2(\mathbf{r}) &= \frac{\mathcal{N}_{0l_2}\mathcal{N}_{0l_0}^2}{w_2}\left(\frac{2r^2}{w_2^2}\right)^{\frac{|l_2|}{2}}\left(\frac{2r^2}{w^2}\right)^{|l_0|}e^{-\frac{r^2}{w_1^2}+il_1\phi} \\ &= \frac{\mathcal{N}_{0l_2}\mathcal{N}_{0l_0}^2\,w_1^{2|l_0|+|l_2|}}{w_2^{|l_2|+1}w^{2|l_0|}}\left(\frac{2r^2}{w_1^2}\right)^{\frac{|l_1|}{2}}\left(\frac{2r^2}{w_1^2}\right)^{p_{\max}} \\ &\quad\times e^{-\frac{r^2}{w_1^2}+il_1\phi}, \end{aligned} \tag{42}$$

where we have defined

$$\begin{aligned} p_{\max} &= |l_0|+\frac{|l_2|-|2l_0-l_2|}{2} \\ &= \begin{cases} 0, & l_0l_2\leq 0 \\ \min(2|l_0|,|l_2|), & l_0l_2>0 \end{cases}. \end{aligned} \tag{43}$$

Then, by expanding the monomial that is raised to the power of $p_{\max}$ in terms of Laguerre polynomials, we get

$$v_2^*(\mathbf{r})u^2(\mathbf{r}) = \sum_{p=0}^{p_{\max}} C_p\, LG_{pl_1}(\mathbf{r};w_1)\,, \tag{44}$$

where

$$C_p = \frac{p_{\max}!\,\mathcal{N}_{0l_2}\mathcal{N}_{0l_0}^2\,w_1^{2|l_0|+|l_2|+1}}{\mathcal{N}_{pl_1}w_2^{|l_2|+1}w^{2|l_0|}}\binom{p_{\max}+|l_1|}{p_{\max}-p}. \tag{45}$$

Therefore, setting $v_1 = LG_{pl_1}(\mathbf{r};w_1)$, which is orthogonal to v_2, and considering the two mode subsystem formed by v_1,v_2, we get an optimal Duan violation of

$$D_{\mathrm{opt}} = \begin{cases} 1-2C_p|Z|+\mathcal{O}(Z^2), & p\leq p_{\max} \\ 1+\mathcal{O}(Z^2), & p>p_{\max} \end{cases}. \tag{46}$$

For $0 \le p \le p_{\max}$, the coefficients C_p are positive with the chosen LG phase convention. Their linear contribution therefore dominates the $\mathcal{O}(Z^2)$ remainder for sufficiently small nonzero Z, establishing entanglement of each corresponding pair. For $p > p_{\max}$, the linear coefficient vanishes: Eq. (46) does not decide whether higher-order correlations produce entanglement. The finite radial support is a property of this first-order coefficient in the specified waist basis.

Throughout the subsequent analysis, we set the reference-mode waist equal to the pump waist, $w_2 = w$. The matching condition $1/w_1^2 = 2/w^2 + 1/w_2^2$ then fixes the partner-mode waist to $w_1 = w/\sqrt{3}$.

Figure 1 illustrates the mode selection for $l_0 = 2$ and $l_2 = 1$, for which $l_1 = 3$ and $p_{\max} = 1$. This is the same input and mode family used in the multimode witness analysis that follows.

B. Consequences for multipartite certification

Let $\mathcal{A}$ denote the reference mode v_2, and let $\mathcal{B}_p$ denote $LG_{p,l_1}(w_1)$ for $p = 0, \ldots, p_{\max}$. These are mutually orthonormal mode parties. Since their number is finite, Eq. (46) gives a common small-distance interval in which every reduced pair $\mathcal{A}|\mathcal{B}_p$ is entangled. Every nontrivial bipartition of the selected set separates $\mathcal{A}$ from at least one partner. The crossing-pair criterion of Sec. III D therefore establishes full inseparability of this set. This applies to its generally mixed reduced state and does not, by itself, certify GME.

For the ideal closed, lossless model, purity of the complete field gives a further consequence. Denote the remaining field modes by $\mathcal{R}$. For $p_{\max} \ge 1$, the grouping

$$\mathcal{A} \mid \mathcal{B}_0 \mid \cdots \mid \mathcal{B}_{p_{\max}-1} \mid (\mathcal{B}_{p_{\max}}\mathcal{R}), \tag{47}$$

includes all degrees of freedom and is pure. The same pairs cross every bipartition, so these $p_{\max} + 2$ grouped parties are genuinely multipartite entangled. For $p_{\max} = 0$, the statement reduces to bipartite entanglement across $\mathcal{A}|(\mathcal{B}_0\mathcal{R})$. This party count is a guaranteed grouping at the specified detection waists, not an upper bound on the number of entangled field modes. It also does not establish GME of the selected single modes after $\mathcal{R}$ is discarded.

We next ask how the certification pattern changes at finite distance and whether GME can be certified directly in the reduced selected state.

VI. ENTANGLEMENT AT FINITE PROPAGATION DISTANCES

A. Diffraction-free Bogoliubov solution

The first-order Duan expression describes the initial generation of correlations but cannot be extrapolated to finite distances: Eq. (46) may become negative for large enough $|Z|$. For finite-distance predictions, we return to the full equation for the same decomposition $\mathcal{E} = \alpha_c + \eta$ and adopt a different approximation: we neglect diffraction and discard the nonlinear fluctuation terms Ω in Eq. (28), while retaining the full dependence on propagation distance. This linearization of the fluctuation dynamics is the Bogoliubov approximation, used to describe quantum fluctuations in paraxial propagation through Kerr media [44]. Appendix A solves the diffraction-free Bogoliubov equations without a Taylor expansion in distance. The dimensionless classical field is then

$$u_Z(\mathbf{r}) = u(\mathbf{r}) e^{-iZ|u(\mathbf{r})|^2}, \tag{48}$$

and the mode correlations are

$$\begin{aligned} f(v_1, v_2) = & -iZ(v_1, v_2; u_Z, u_Z) \\ & - Z^2(v_1, v_2, u_Z; u_Z, u_Z, u_Z), \\ h(v_1, v_2) = & Z^2(v_1, u_Z, u_Z; v_2, u_Z, u_Z). \end{aligned} \tag{49}$$

These expressions agree with the preceding first-order result and are exact within the diffraction-free linearized fluctuation equations. Their use at finite distance requires negligible diffraction over the interaction region, which is a reasonable approximation for softly focused beams, as well as a small fluctuation backreaction. The numerical results below use these correlations under those assumptions.

B. Two-mode certification

Figure 2 compares the phase-optimized Duan witness at fixed $l_0 = 2$ for $l_2 = -1, 0, 1, 3$. The corresponding first-order radial cutoffs are $p_{\max} = 0, 0, 1, 3$, respectively. The first two panels illustrate the single radial partner selected for opposite-sign or zero reference-mode OAM, while the latter two illustrate the larger radial support available for equal-sign OAM. In particular, all four plotted radial indices lie within the first-order support for $l_2 = 3$. The $l_2 = 1$ panel uses the same input and radial partners as the schematic in Fig. 1 and the multimode witness analysis below.

C. Entanglement across multimode bipartitions

Now we turn to more parties. We set $l_0 = 2$ and $l_2 = 1$, so that $l_1 = 2l_0 - l_2 = 3$ and $p_{max} = \min(2|l_0|, |l_2|) = 1$. Furthermore, we define the modes as $\mathcal{A} = LG_{0,1}(w_2 = w)$, $\mathcal{B}_0 = LG_{0,3}(w_1)$, $\mathcal{B}_1 = LG_{1,3}(w_1)$ and $\mathcal{C} = LG_{2,3}(w_1)$. Thus $\mathcal{C}$ is a radial mode outside the first-order support for coupling to $\mathcal{A}$. The short-distance calculation certifies the pairs $\mathcal{A}|\mathcal{B}_0$ and $\mathcal{A}|\mathcal{B}_1$, while it is inconclusive for $\mathcal{A}|\mathcal{C}$. By Sec. V B, these two pairs already certify six of the seven bipartitions; the remaining cut is $\mathcal{A}\mathcal{B}_0\mathcal{B}_1|\mathcal{C}$. In Fig. 3 we apply FullyWit to the reduced

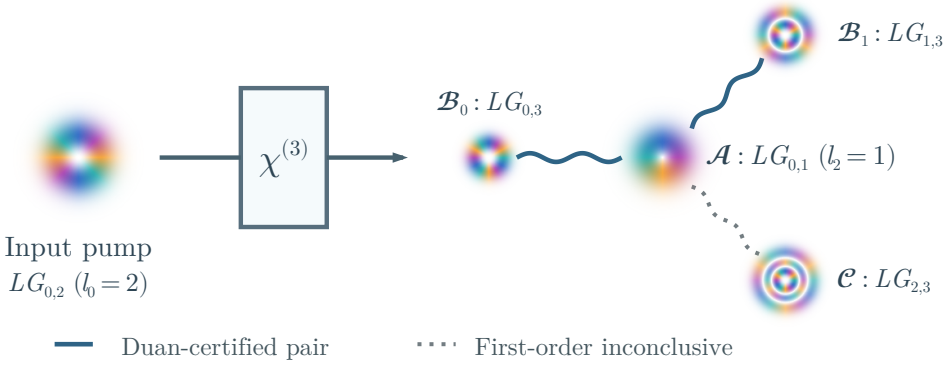


FIG. 1. Schematic of radial–angular mode coupling and pairwise certification for an OAM pump with $l_0 = 2$. The nonlinear medium generates an output field containing the reference mode $\mathcal{A} = LG_{0,1}(w_2 = w)$ and radial partners $\mathcal{B}_0 = LG_{0,3}(w_1)$, $\mathcal{B}_1 = LG_{1,3}(w_1)$, and $\mathcal{C} = LG_{2,3}(w_1)$. Here $l_2 = 1$, $l_1 = 3$, and $p_{\max} = 1$. Solid links indicate the first-order Duan-violating pairs $\mathcal{A}|\mathcal{B}_p$ for $p = 0, 1$; the dashed link marks $\mathcal{A}|\mathcal{C}$ as inconclusive at first order, not as unentangled. Phase is encoded by hue and intensity by transparency in the mode profiles.

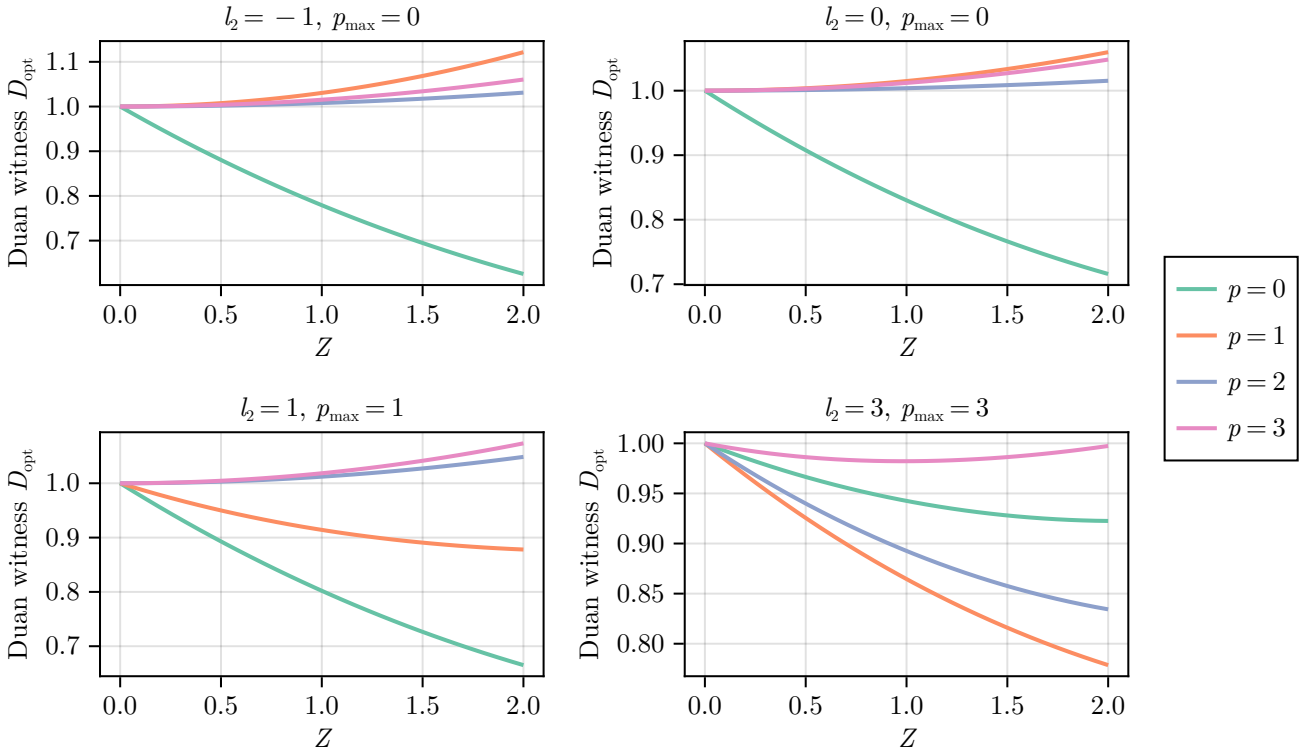


FIG. 2. Phase-optimized Duan witness D_{opt} for the diffraction-free Bogoliubov solution with fixed pump OAM $l_0 = 2$ and reference-mode OAM $l_2 = -1, 0, 1, 3$. Curves correspond to radial indices $p = 0, \ldots, 3$ of the partner mode, with $l_1 = 4 - l_2$, $w_1 = w/\sqrt{3}$, and $w_2 = w$. Values below one certify two-mode entanglement.

covariance matrix of these four modes and optimize a witness separately for each bipartition. The numerical curves form three groups:

- $\mathcal{AB}_1\mathcal{C}|\mathcal{B}_0$, $\mathcal{AB}_1|\mathcal{B}_0\mathcal{C}$, $\mathcal{AC}|\mathcal{B}_0\mathcal{B}_1$ and $\mathcal{A}|\mathcal{B}_0\mathcal{B}_1\mathcal{C}$ show the lowest witness values and hence the largest margins below the separability threshold. These share

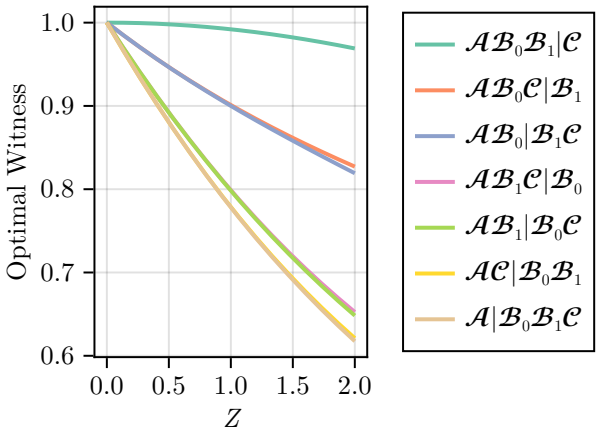


FIG. 3. Optimized covariance witnesses for the seven bipartitions of the reduced four-mode state $\mathcal{AB}_0\mathcal{B}_1\mathcal{C}$. Values below one certify entanglement across the corresponding cut. Here $l_0 = 2$ and $l_2 = 1$, so that $l_1 = 2l_0 - l_2 = 3$ and $p_{max} = \min(2|l_0|, |l_2|) = 1$.

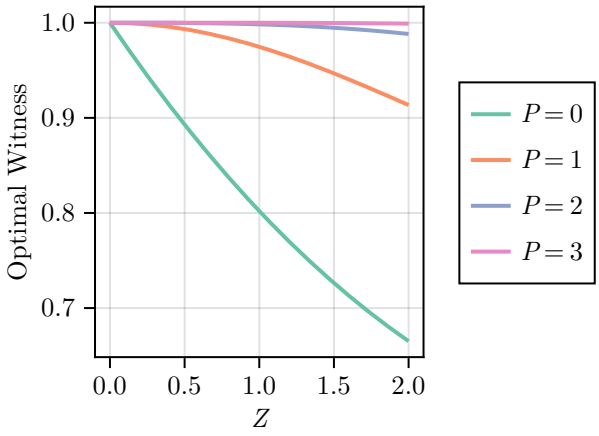


FIG. 4. Optimized witnesses for the reduced selected sets containing $\mathcal{A}$ and radial partners $0 \le p \le P$. Values below one certify bipartite entanglement for $P = 0$ and genuine multipartite entanglement for $P \ge 1$. Here $l_0 = 2$ and $l_2 = 1$, so that $l_1 = 2l_0 - l_2 = 3$ and $p_{max} = \min(2|l_0|, |l_2|) = 1$.

the fact that $\mathcal{A}$ and $\mathcal{B}_0$ belong to different parties, giving a stronger second-moment entanglement certification across these partitions.

- $\mathcal{AB}_0\mathcal{C}|\mathcal{B}_1$ and $\mathcal{AB}_0|\mathcal{B}_1\mathcal{C}$ show intermediate witness values, still maintaining a linear start. Here the entangled pair $\mathcal{A}|\mathcal{B}_1$ crosses the cut, while $\mathcal{A}$ and $\mathcal{B}_0$ belong to the same party.

- $\mathcal{AB}_0\mathcal{B}_1|\mathcal{C}$ shows the smallest margin below the separability threshold, with a numerical onset consistent with a quadratic fall. Here both first-order entangled pairs lie within $\mathcal{AB}_0\mathcal{B}_1$, so the individual-pair argument gives no first-order certificate across this cut.

The crossing-pair argument explains why the first six cuts admit linear violations, but does not predict the optimized witness values. The small violation for $\mathcal{AB}_0\mathcal{B}_1|\mathcal{C}$ illustrates that modes outside the first-order radial support can nevertheless participate in entanglement at finite distance.

D. Genuine multipartite entanglement of selected modes

Finally, we apply the biseparable-class optimization of Sec. III D to the reduced selected modes. Unlike testing each cut separately, a MultiWit violation excludes all mixtures in Eq. (16) and therefore certifies GME after the remaining field has been discarded.

In Fig. 4, we show the witness for sets of the form $\{LG_{0,1}(w_2 = w)\} \cup \{LG_{p,3}(w_1) \mid p \le P\}$, with P varying from 0 to 3, while maintaining $l_0 = 2$. The $P = 0$ case is bipartite; for $P \ge 1$, a violation certifies genuine multipartite entanglement. The numerical results display a pattern related to the first-order radial support: the sets with $P \le p_{\max} = 1$ have lower witness values and hence larger margins below the separability threshold.

In particular, for $P = 1$ Eq. (47) certifies the pure-state grouping $\mathcal{A}|\mathcal{B}_0|(\mathcal{B}_1\mathcal{R})$, whereas a MultiWit violation certifies the reduced three-mode state on $\mathcal{A}|\mathcal{B}_0|\mathcal{B}_1$. The latter shows that genuine multipartite entanglement survives discarding the remaining field modes. Violations for larger selected sets, when present, involve modes outside the first-order support and are finite-distance results within the Bogoliubov approximation.

VII. EXPERIMENTAL IMPLEMENTATION

Figure 5 outlines two complementary implementations of the proposed entanglement scheme: local characterization of the covariance matrix and spatial distribution of the correlated modes. In both cases, a spatial light modulator (SLM) prepares an LG pump that propagates through a $\chi^{(3)}$ nonlinear medium. The interaction generates correlations between transverse modes, which are subsequently probed by balanced homodyne detection with a spatially shaped local oscillator (LO).

For this preparation stage, the required interaction strength can be assessed through the accumulated nonlinear phase. Fontaine et al. [45] investigated a propagating fluid of light in a hot rubidium vapor cell of length $L = 7.5\,\text{cm}$ at a wavelength $\lambda = 780\,\text{nm}$, finding a nonlinear index change proportional to the intensity, consistent with an unsaturated Kerr response. Taking $|\Delta n| \simeq 5 \times 10^{-6}$ as a representative value within their

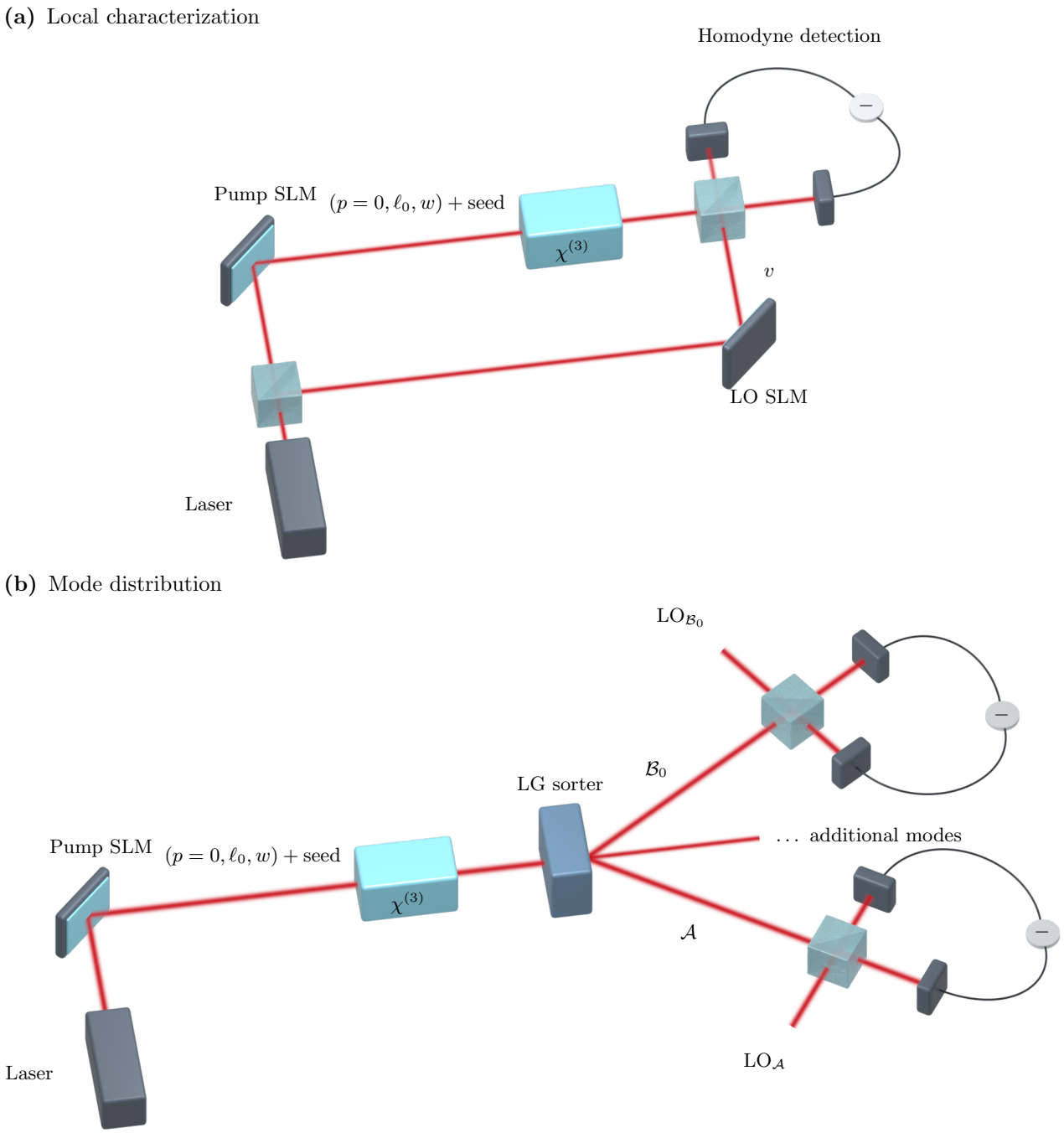


FIG. 5. Experimental implementation schemes. (a) Local characterization by balanced homodyne detection with a spatially shaped local oscillator in mode v. (b) Mode distribution using a Laguerre–Gaussian (LG) mode sorter, followed by balanced homodyne detection of the separated modes with their respective local oscillators. In both schemes, a spatial light modulator (SLM) shapes the pump before the $\chi^{(3)}$ nonlinear medium. LO denotes the local oscillator.

experimental range gives the effective phase estimate

$$\Phi_{\mathrm{NL}} \simeq \frac{2\pi}{\lambda} L|\Delta n| \simeq 3.0\,\mathrm{rad}. \tag{50}$$

In our diffraction-free model, the local phase magnitude is $|Z||u(\mathbf{r})|^2$. For an $LG_{0,l}$ pump, its peak value is

$$\Phi_{\mathrm{pk}} = |Z|\frac{2}{\pi}\frac{|l|^{|l|}e^{-|l|}}{|l|!}, \tag{51}$$

where the last fraction is unity for $l = 0$. At $|Z| = 1$, the peak phases are approximately 0.64, 0.23, and 0.17 rad

for $|l| = 0$, 1, and 2, respectively. Their smallness relative to the experimental phase scale supports the accessibility of $|Z| = 1$, where our finite-distance calculations already predict entanglement signatures.

Transferring this estimate to the proposed setup requires accounting for the power delivered to the medium after LG preparation and filtering, as well as the change in peak intensity associated with the selected spatial profile. Preparation losses can in principle be compensated by a more powerful input laser: for an efficiency η, maintaining the same power at the cell requires increasing the source power by a factor $1/\eta$. The comparison therefore provides an estimate of a feasible interaction strength, while the observable entanglement also depends on propagation and detection losses and noise from the medium.

To characterize the resulting state locally, the arrangement in Fig. 5(a) uses a Mach–Zehnder interferometer. One arm contains the pump preparation and nonlinear medium, while the other supplies the LO, shaped by a second SLM into a configurable mode v. Balanced homodyne detection measures $\Gamma(v, v)$, giving the diagonal covariance elements. The off-diagonal elements follow from

$$\begin{aligned}\Gamma(v_1, v_2) =& \Gamma\left(\frac{v_1 + v_2}{\sqrt{2}}, \frac{v_1 + v_2}{\sqrt{2}}\right) \\ &- \frac{1}{2}\left[\Gamma(v_1, v_1) + \Gamma(v_2, v_2)\right],\end{aligned} \tag{52}$$

which requires an additional measurement with the LO prepared in the superposition $(v_1 + v_2)/\sqrt{2}$. These measurements provide the covariance information needed to evaluate the entanglement criteria discussed above.

Coherent seeds collinear with the pump can assist the phase control required for these measurements by populating the selected modes $\mathcal{A}, \mathcal{B}_0, \ldots$. Appendix B examines this possibility and shows that, for the configurations considered, a seed fraction of 1% of the total input power produces only modest changes in the finite-distance witness curves. The usefulness of the resulting coherent references depends on the experimental sensitivity.

Beyond local characterization, the correlated modes can be distributed to separate parties using the arrangement in Fig. 5(b). An LG mode sorter [46–50] spatially separates the output components, allowing each party to perform balanced homodyne detection with an appropriate LO. The two arrangements thus address both the covariance measurements needed to certify the generated entanglement and the spatial separation needed to make its modes individually accessible.

VIII. CONCLUSION

We have derived a chiral selection rule for the first-order anomalous correlations generated by self-phase modulation of a coherent OAM beam. At specified detection waists, radial–angular coupling fixes a finite set of partner modes with nonzero linear Duan violations. These violations guarantee pairwise entanglement and full inseparability of the selected set at sufficiently short nonzero distances. For the full pure field, assigning the remaining modes to one selected party also gives a multipartite grouping whose entanglement follows from the same correlations.

Finite-distance calculations within the diffraction-free Bogoliubov approximation show that the first-order support remains useful for organizing entanglement certification. In the configurations investigated, optimized witnesses reveal weaker violations involving modes outside that support, so it is not an exact finite-distance entanglement boundary. Witnesses valid for all biseparable states additionally certify GME in reduced sets of detection modes. These conclusions concern the certification margins of the chosen criteria and do not quantify the amount of entanglement.

The results identify OAM and the detection profiles as controls over the radial mode combinations that admit covariance-based entanglement certification. They can be investigated in four-wave mixing and self-phase modulation settings where diffraction over the interaction region and fluctuation backreaction are sufficiently small. These findings can be useful to quantum information protocols that require control over multipartite entanglement.

Appendix A: Diffraction-free Bogoliubov evolution

The first-order calculation in the main text retains diffraction and all nonlinear fluctuation terms in Eq. (28). Here we start from the same full fluctuation equation and adopt a different approximation: we neglect diffraction over the interaction region and discard the quadratic and cubic fluctuation terms Ω of Eq. (29), while retaining the full dependence on propagation distance. The resulting Bogoliubov equations are solved without expanding in the propagation distance or the accumulated classical nonlinear phase.

With the same decomposition $\mathcal{E} = \alpha_c + \eta$ and input $\alpha(\mathbf{r}) = \alpha_c(\mathbf{r}, 0)$, these equations are

$$\begin{aligned}2ik\partial_z\alpha_c &= g|\alpha_c|^2\alpha_c, \\ 2ik\partial_z\eta &= g\left(2|\alpha_c|^2\eta + \alpha_c^2\eta^\dagger\right).\end{aligned} \tag{A1}$$

The classical intensity is constant along propagation. Defining

$$I(\mathbf{r}) = |\alpha(\mathbf{r})|^2, \qquad \theta(\mathbf{r}, z) = \frac{gz}{2k}I(\mathbf{r}), \tag{A2}$$

we have

$$\alpha_c(\mathbf{r}, z) = \alpha(\mathbf{r})e^{-i\theta(\mathbf{r},z)}. \tag{A3}$$

1. Local fluctuation solution

Each transverse position evolves independently. Introduce the rotating-frame fluctuation $b(\mathbf{r}, z) =$

$e^{i\theta(\mathbf{r},z)}\eta(\mathbf{r},z)$. Suppressing the position argument temporarily, Eq. (A1) becomes

$$2ik\partial_z b = g\left(Ib + \alpha^2 b^\dagger\right). \tag{A4}$$

Together with its conjugate, this gives

$$2ik\partial_z \begin{pmatrix} b \\ b^\dagger \end{pmatrix} = gM \begin{pmatrix} b \\ b^\dagger \end{pmatrix}, \qquad M = \begin{pmatrix} I & \alpha^2 \\ -\alpha^{*2} & -I \end{pmatrix}. \tag{A5}$$

The matrix is independent of z and satisfies $M^2 = 0$. Therefore its exponential terminates exactly:

$$\exp\left(-\frac{igz}{2k}M\right) = \mathbb{I} - \frac{igz}{2k}M. \tag{A6}$$

Returning to η and writing $\eta_0(\mathbf{r}) = \eta(\mathbf{r},0)$, we obtain

$$\eta(\mathbf{r},z) = e^{-i\theta}\Big[(1-i\theta)\eta_0(\mathbf{r}) - i\frac{gz}{2k}\alpha^2(\mathbf{r})\eta_0^\dagger(\mathbf{r})\Big]. \tag{A7}$$

Although the rotating-frame coefficients are linear in z, this is not a first-order Taylor approximation. It is the exact solution of the diffraction-free linearized fluctuation equations.

Writing $\eta = A\eta_0 + B\eta_0^\dagger$, the coefficients are

$$A = e^{-i\theta}(1-i\theta), \qquad B = -i\frac{gz}{2k}\alpha^2 e^{-i\theta}. \tag{A8}$$

They obey $|A|^2 - |B|^2 = 1$, so the field commutator is preserved. For the coherent input, $\langle\eta_0\rangle = 0$ and hence $\langle\eta(z)\rangle = 0$. Thus the mean field equals α_c within this approximation.

2. Field and mode correlations

The initial fluctuations satisfy

$$\begin{aligned}\langle\eta_0(\mathbf{r}_1)\eta_0^\dagger(\mathbf{r}_2)\rangle &= \delta^{(2)}(\mathbf{r}_1-\mathbf{r}_2),\\ \langle\eta_0^\dagger(\mathbf{r}_2)\eta_0(\mathbf{r}_1)\rangle &= \langle\eta_0(\mathbf{r}_1)\eta_0(\mathbf{r}_2)\rangle = 0.\end{aligned} \tag{A9}$$

Consequently the correlations are

$$\begin{aligned}F(\mathbf{r}_1,\mathbf{r}_2;z) &= A(\mathbf{r}_1)B(\mathbf{r}_2)\delta^{(2)}(\mathbf{r}_1-\mathbf{r}_2),\\ H(\mathbf{r}_1,\mathbf{r}_2;z) &= B(\mathbf{r}_1)B^*(\mathbf{r}_2)\delta^{(2)}(\mathbf{r}_1-\mathbf{r}_2).\end{aligned} \tag{A10}$$

Substituting the coefficients gives

$$\begin{aligned}F(\mathbf{r}_1,\mathbf{r}_2;z) &= \left[-\frac{igz}{2k} - \left(\frac{gz}{2k}\right)^2 I(\mathbf{r}_1)\right] \times \alpha_c^2(\mathbf{r}_1,z)\delta^{(2)}(\mathbf{r}_1-\mathbf{r}_2),\\ H(\mathbf{r}_1,\mathbf{r}_2;z) &= \left(\frac{gz}{2k}\right)^2 I^2(\mathbf{r}_1)\delta^{(2)}(\mathbf{r}_1-\mathbf{r}_2).\end{aligned} \tag{A11}$$

Using the main-text scaling, $\theta = Z|u|^2$ and $u_Z = u\exp(-iZ|u|^2)$, projection onto the detection modes yields

$$\begin{aligned}f(v_1,v_2) &= -iZ(v_1,v_2;u_Z,u_Z) - Z^2(v_1,v_2,u_Z;u_Z,u_Z,u_Z),\\ h(v_1,v_2) &= Z^2(v_1,u_Z,u_Z;v_2,u_Z,u_Z).\end{aligned} \tag{A12}$$

The phases in h cancel, whereas those in f retain the nonlinear classical evolution. Expanding Eq. (A12) at small Z recovers $f = -iZ(v_1,v_2;u,u) + \mathcal{O}(Z^2)$ and $h = \mathcal{O}(Z^2)$, in agreement with the first-order calculation in the main text. At finite distance, these expressions are exact only within the diffraction-free Bogoliubov approximation, not for the full quantum Kerr equation.

The approximation requires negligible diffraction of the relevant transverse structures over the interaction length, as well as sufficiently small fluctuation interactions and backreaction on the classical field. A large pump Rayleigh range is a useful initial scale, but finer structure generated by the nonlinear phase can have shorter diffraction lengths. The derivation applies to any coherent input profile, including a coherent pump plus seeds.

The Bogoliubov approximation also permits retaining diffraction in both the classical background and the linear fluctuation equations. Quantum correlations can then be computed numerically by propagating the corresponding Green functions [51, 52], or by back-propagating the detection functions with the adjoint fluctuation equations [53, 54] and evaluating their correlations in the known input state. Such a numerical treatment with diffraction lies beyond the scope of the present work, which uses the analytical diffraction-free solution above.

Appendix B: Coherent seeding of the selected modes

The homodyne schemes of Sec. VII can benefit from coherent references in all selected detection modes for alignment and phase control. An unseeded pump with OAM l_0 has no coherent mean in the detected OAM families $l_1 + l_2 = 2l_0$, with $l_2 \neq l_0$, although these modes can be correlated and measured with an external local oscillator. Here we add weak coherent seeds directly to the reference mode and every radial partner selected by the first-order rule, and assess their effect on the entanglement witnesses.

We retain the mode definitions of Sec. V: the pump profile is $LG_{0,l_0}(\mathbf{r};w)$, the reference mode is $v_2 = LG_{0,l_2}(\mathbf{r};w_2)$, and a radial partner is $v_1 = LG_{p,l_1}(\mathbf{r};w_1)$, with

$$l_1 = 2l_0 - l_2, \qquad \frac{1}{w_1^2} = \frac{2}{w^2} + \frac{1}{w_2^2}. \tag{B1}$$

We seed the reference mode $\mathcal{A}$ and all radial partners $\mathcal{B}_p$, $p = 0,\ldots,p_{\max}$, with the cutoff defined in Eq. (43).

Choosing equal amplitudes and phases gives the normalized seed profile

$$s = \frac{v_2 + \sum_{p=0}^{p_{\max}} LG_{p,l_1}(\mathbf{r}; w_1)}{\sqrt{N_s}}, \qquad N_s = p_{\max} + 2. \quad \text{(B2)}$$

The seed modes are mutually orthogonal and orthogonal to the pump. For mutually coherent, collinear fields with the same frequency, polarization, and temporal mode, we therefore write

$$\alpha(\mathbf{r}) = \|\alpha\| \left[\sqrt{1-\epsilon^2}\, LG_{0,l_0}(\mathbf{r}; w) + \epsilon\, s(\mathbf{r})\right], \quad \text{(B3)}$$

where $0 \le \epsilon < 1$ is the seed amplitude parameter and ϵ^2 is the fraction of the total input power assigned to the seed. Each selected mode then has the input mean amplitude $\|\alpha\|\epsilon/\sqrt{N_s}$. We hold the total occupation $\|\alpha\|^2$ fixed, so the comparison with the unseeded case uses the same propagation coordinate,

$$u = w\left[\sqrt{1-\epsilon^2}\, LG_{0,l_0}(\mathbf{r}; w) + \epsilon\, s\right], \qquad Z = \frac{gz\|\alpha\|^2}{2kw^2}. \quad \text{(B4)}$$

The leading change in the selected correlations follows directly from their nonlinear overlap. Since neither seed family has OAM l_0, the pump–seed terms in u^2 cannot contribute to the channel $l_1 + l_2 = 2l_0$. The two seed families do have complementary OAM, however, and their product contributes at order ϵ^2. For a detected partner $v_1 = LG_{p,l_1}(\mathbf{r}; w_1)$, let C_p denote the unseeded coefficient defined in the main text. Its seeded counterpart is

$$\begin{aligned} C_p(\epsilon) &= (v_1, v_2; u, u) \\ &= (1-\epsilon^2) C_p + \frac{2\epsilon^2}{N_s} \sum_{p'=0}^{p_{\max}} K_{pp'}, \\ K_{pp'} &= w^2\big(v_1, v_2; v_2, LG_{p',l_1}(w_1)\big). \end{aligned} \quad \text{(B5)}$$

Here p' labels the seeded radial partners, and $K_{pp'}$ uses the spatial overlap notation of the main text, with spatial arguments suppressed. The first-order anomalous correlation and phase-optimized Duan witness are consequently

$$\begin{aligned} f(v_1, v_2) &= -iZC_p(\epsilon) + \mathcal{O}(Z^2), \\ D_{\text{opt}} &= 1 - 2|Z|\,|C_p(\epsilon)| + \mathcal{O}(Z^2). \end{aligned} \quad \text{(B6)}$$

Thus the selected overlap coefficients change by $\mathcal{O}(\epsilon^2)$, with no correction linear in ϵ. These changes arise because the seed participates in the nonlinear interaction; a coherent displacement alone does not change centered correlations. The weighted overlaps $K_{pp'}$ need not vanish beyond the original radial cutoff, so weak seeding preserves the dominant correlation pattern without preserving every first-order zero exactly. Small absolute corrections can also matter for pairs close to a witness threshold.

To assess the effect at finite distance, we insert Eq. (B4) into the diffraction-free Bogoliubov correlation formulas of Appendix A. Figure 6 uses the same pump and detection parameters as Fig. 2, with $w_2 = w$, $w_1 = w/\sqrt{3}$, and $\epsilon = 0.1$: one percent of the total input power is shared equally among all N_s selected modes. Every curve within a panel describes this same seeded input state, including curves for unseeded radial partners outside the first-order support.

Compared with the unseeded results in Fig. 2, weak seeding leaves the overall certification pattern largely unchanged. Pairs with substantial unseeded violations retain clear violations, and the relative behavior of the radial curves remains similar. The small shifts can nevertheless affect pairs close to the separability threshold. For example, the $l_2 = 3$, $p = 3$ pair ceases to violate the witness near the end of the plotted interval, although it remains certified at shorter distances. Failure to violate the witness there is inconclusive about entanglement.

These examples show that direct coherent references can be introduced simultaneously into all modes selected by the first-order rule with only modest changes to the finite-distance witness curves for the seed strength and phases considered. This agrees qualitatively with the weak-seed analysis above, while the near-threshold changes emphasize that preserving the dominant pattern does not imply an exact finite-distance selection rule.

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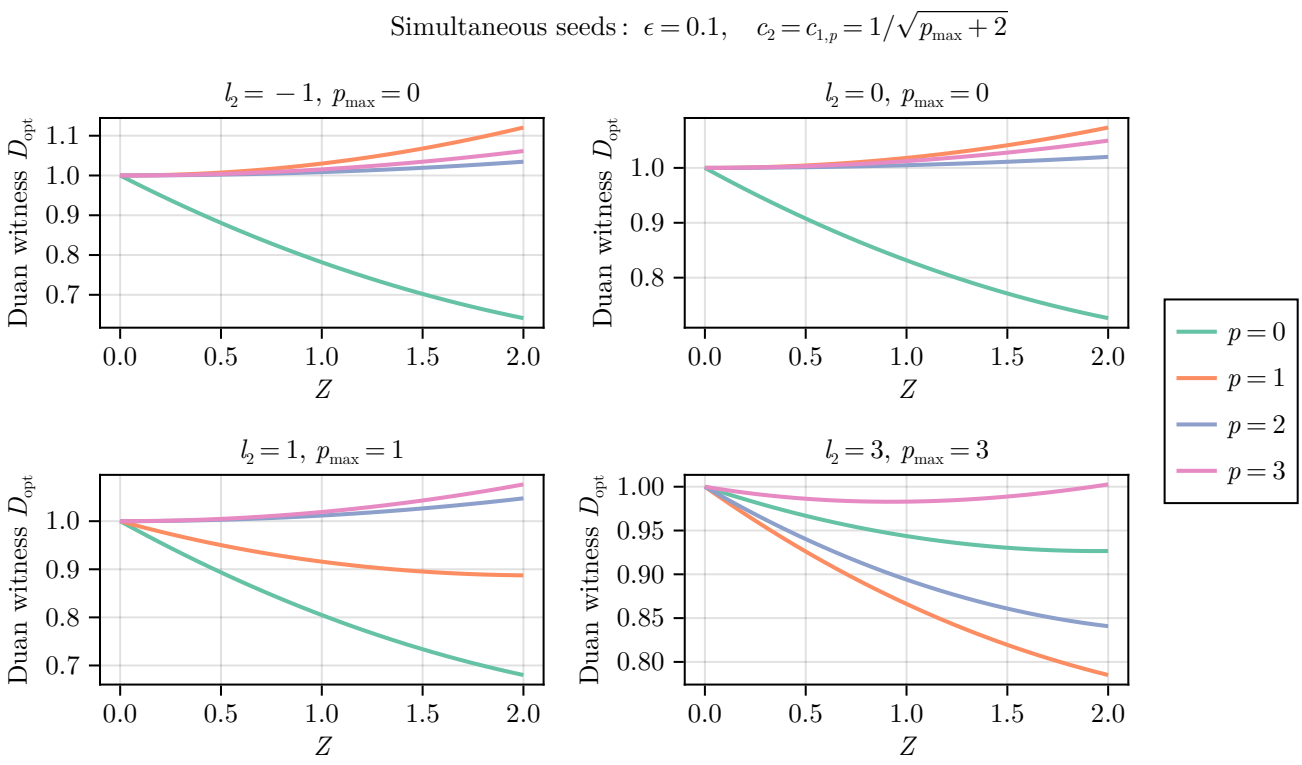


FIG. 6. Phase-optimized Duan witness with simultaneous coherent seeding of the reference mode and all radial partners $p = 0, \ldots, p_{\max}$, with seed amplitude $\epsilon = 0.1$ (power fraction $\epsilon^2 = 0.01$) and equal amplitudes and phases as in Eq. (B2). The pump OAM is $l_0 = 2$, and the detection modes and waists match Fig. 2. All curves within each panel use the same seeded input. Values below one certify two-mode entanglement within the diffraction-free Bogoliubov model.

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